Tutorial 5: Convolution Neural Network

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1. Introduction

The objective of this project is to build and train a deep learning model capable of identifying objects across 10 different classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). The model is implemented using **PyTorch**.

2. Dataset and Preprocessing

The **CIFAR-10** dataset was used, consisting of 60,000 32x32 color images.

- **Normalization:** Pixel values were normalized to a range of [-1, 1] using a mean of 0.5 and a standard deviation of 0.5 for all three color channels (RGB) to improve model convergence.
- **Batching:** Data was loaded in batches of 64 using PyTorch DataLoaders.

3. Model Architecture

The model follows a classic CNN structure with three convolutional layers for feature extraction and two fully connected (Dense) layers for classification.

Layer Type	Configuration	Activation
Convolution 1	3 input channels, 32 filters, 3x3 kernel	ReLU
Max Pooling	2x2 pool size	-
Convolution 2	32 input, 64 filters, 3x3 kernel	ReLU
Max Pooling	2x2 pool size	-

Layer Type	Configuration	Activation
Convolution 3	64 input, 64 filters, 3x3 kernel	ReLU
Fully Connected 1	1024 input features → 64 neurons	ReLU
Fully Connected 2	64 input features → 10 output classes	Softmax (via Loss)

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4. Training Process

The model was trained using the following parameters:

- **Optimizer:** Adam (Learning Rate: 0.001)
- **Loss Function:** Cross-Entropy Loss
- **Epochs:** 10

Training Logs

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Epoch 1, Loss: 1.5440, Accuracy: 43.55%
Epoch 2, Loss: 1.2050, Accuracy: 57.06%
Epoch 3, Loss: 1.0359, Accuracy: 63.52%
Epoch 4, Loss: 0.9233, Accuracy: 67.69%
Epoch 5, Loss: 0.8400, Accuracy: 70.36%
Epoch 6, Loss: 0.7732, Accuracy: 72.94%
Epoch 7, Loss: 0.7166, Accuracy: 75.00%
Epoch 8, Loss: 0.6668, Accuracy: 76.71%
Epoch 9, Loss: 0.6231, Accuracy: 78.23%
Epoch 10, Loss: 0.5859, Accuracy: 79.59%
```

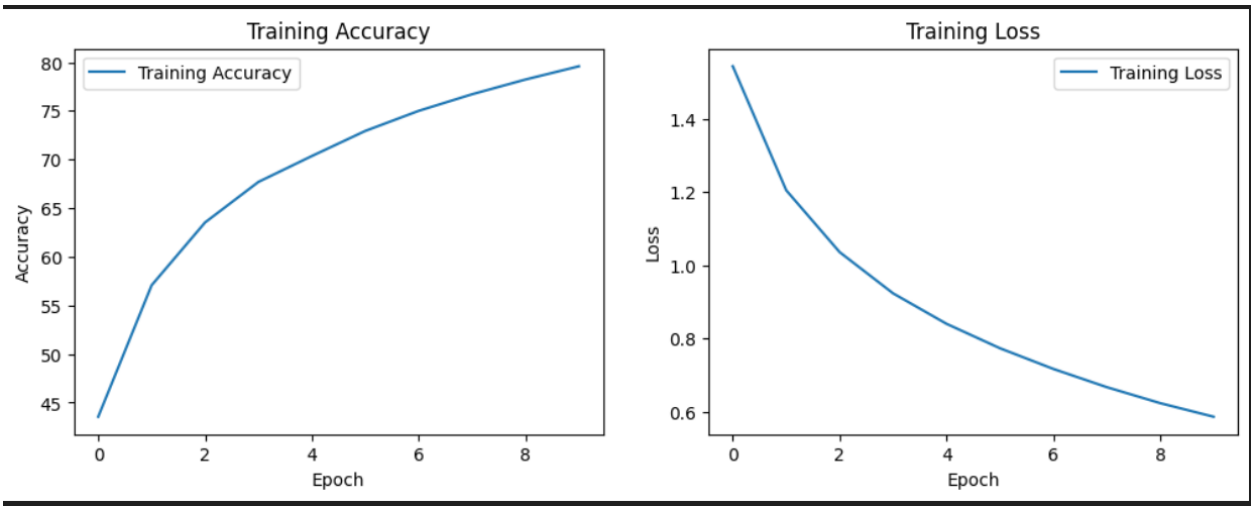
5. Results and Performance

The model showed consistent improvement over 10 epochs.

- **Starting Accuracy:** ~43.55% (Epoch 1)
- **Final Training Accuracy: 79.59%** (Epoch 10)
- **Final Training Loss:** 0.5859

Performance Visualization

The learning curves indicate that the model effectively learned the features of the dataset, with loss decreasing and accuracy increasing steadily.



6. Conclusion

The CNN successfully achieved nearly 80% accuracy on the training set within 10 epochs. The steady decline in loss suggests the learning rate was appropriate. Future improvements could include:

1. **Data Augmentation:** Adding random crops or flips to prevent overfitting.
2. **Dropout Layers:** To improve generalization on the test set.
3. **Increased Depth:** Adding more filters or layers to capture more complex patterns.